Stochastic Feature Selection Networks for Brain MRI Analysis: Probabilistic Dimensionality Reduction with Uncertainty Quantification

A Gumbel-Softmax Approach to Unsupervised Medical Imaging

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***Abstract*—Traditional autoencoders for brain MRI analysis lack explicit feature selection and uncertainty quantification capabilities. This paper introduces a Stochastic Feature Selection Network (SFSN) using Gumbel-Softmax sampling for unsupervised dimensionality reduction of Alzheimer's MRI data. The architecture combines differentiable stochastic gates for feature selection, multi-objective optimization, and uncertainty estimation. Evaluation on 5,000 brain MRI scans compared SFSN against a deterministic baseline autoencoder. While the baseline achieved lower reconstruction error (0.0046 vs 0.0064), SFSN demonstrated superior manifold preservation with 2.4% improvement in trustworthiness and 2.7% improvement in continuity. The stochastic mechanism selected 49% of input features while providing interpretable importance scores and uncertainty measures. The results demonstrate that probabilistic feature selection improves neighborhood preservation despite higher reconstruction error, offering interpretability and uncertainty awareness essential for medical imaging applications.**

***Keywords*—Stochastic neural networks, feature selection, brain MRI, Alzheimer's disease, dimensionality reduction, uncertainty quantification**

1. Introduction

1.1. Problem Motivation and Background

Alzheimer's disease represents one of the most pressing challenges in modern healthcare, affecting over 55 million people worldwide and standing as the leading cause of dementia. Early detection and monitoring of Alzheimer's progression through neuroimaging has become increasingly crucial, as structural brain changes often precede clinical symptoms by years. Brain MRI scans provide detailed anatomical information that can reveal subtle patterns associated with neurodegeneration, including hippocampal atrophy, ventricular enlargement, and cortical thinning. However, medical imaging analysis, particularly brain MRI scans for Alzheimer's research, generates high-dimensional data that poses significant challenges for automated analysis systems. Traditional dimensionality reduction techniques often fail to capture the complex, non-linear relationships inherent in medical imaging data while simultaneously providing interpretable feature selection mechanisms. The curse of dimensionality in medical imaging not only increases computational complexity but also hampers the ability to identify clinically relevant features that could aid in diagnostic processes. Brain MRI scans contain thousands of pixels, each representing crucial anatomical information about brain structures that may be affected by Alzheimer's pathology. However, not all features contribute equally to meaningful representations of disease-related changes. The challenge lies in automatically identifying and selecting the most informative features while preserving the essential characteristics needed for accurate reconstruction and analysis. This is particularly important in Alzheimer's research, where subtle anatomical changes must be detected against the backdrop of normal aging variations. Furthermore, the inherent uncertainty in medical data arising from acquisition noise, anatomical variability, and disease heterogeneity necessitates approaches that can quantify and model this uncertainty explicitly.

1.2. Choice of Application and Justification

This project focuses on brain MRI analysis using the Alzheimer's MRI Dataset, specifically applying unsupervised learning techniques to ignore diagnostic labels and concentrate on feature representation learning. The choice of brain MRI data is motivated by several factors. **High dimensionality** is an important aspect, since brain MRI scans represent complex 3D structures flattened into high-dimensional vectors (4096 dimensions for 64×64 images), which makes them ideal for testing dimensionality reduction techniques. **Clinical relevance** also plays a major role, as developing better feature representations for brain imaging could have significant implications for automated diagnostic systems and medical image analysis workflows. In addition, the **unsupervised learning context** ensures that by ignoring diagnostic labels, the model learns inherent data structures without bias toward specific clinical outcomes, potentially discovering novel feature patterns. Finally, **uncertainty quantification** is crucial because medical data inherently contains measurement uncertainties and anatomical variations, making uncertainty estimation essential for reliable automated analysis.

1.3. Research Questions and Objectives

The primary research objectives of this project are:

i. Can stochastic feature selection improve dimensionality reduction quality compared to deterministic approaches? Investigating whether probabilistic feature selection mechanisms can identify more meaningful feature subsets than traditional deterministic autoencoders.

ii. How does non-deterministic feature selection affect reconstruction quality and feature interpretability? Examining the trade-offs between reconstruction accuracy and feature selectivity in stochastic versus deterministic models.

iii. Can uncertainty quantification in feature selection provide additional insights into data representation quality? Evaluating whether explicit uncertainty modeling enhances the robustness and interpretability of learned representations.

iv. What is the optimal balance between feature sparsity and reconstruction fidelity? Determining hyperparameter configurations that maximize both feature selectivity and reconstruction quality.

2. Related Work

2.1. Brief Survey of Existing Approaches

Traditional dimensionality reduction techniques for medical imaging have primarily relied on deterministic approaches. Principal Component Analysis (PCA) and its variants have been extensively used but suffer from linear assumptions that poorly capture complex anatomical structures. Deep autoencoders have shown promise in learning non-linear representations but typically lack explicit feature selection mechanisms and uncertainty quantification.

Recent advances in stochastic neural networks have introduced probabilistic elements into deep learning architectures. Variational Autoencoders (VAEs) incorporate stochasticity in the latent space but do not explicitly address feature selection in the input space. Attention mechanisms have been used for implicit feature selection but lack the explicit probabilistic formulation needed for uncertainty quantification.

Gumbel-Softmax and concrete distributions have been applied to discrete feature selection problems, enabling differentiable discrete sampling. However, these approaches have seen limited application in medical imaging contexts, particularly for unsupervised feature selection in high-dimensional image data.

2.2. Limitations of Current Methods

Existing approaches exhibit several limitations. Lack of explicit feature selection is one concern, as most autoencoders learn implicit feature representations without providing interpretable feature importance scores. Another issue is their deterministic nature, since traditional methods lack uncertainty quantification and only provide point estimates without confidence measures. A further limitation comes from the fixed architecture, where most approaches rely on fixed encoder-decoder designs without adaptive feature selection capabilities. Finally, there is the problem of limited medical imaging application, because few studies have specifically addressed the unique challenges of medical imaging data, particularly its inherent noise and anatomical variations.

2.3. Novelty of Your Approach

This project introduces several novel contributions. The first is the **Stochastic Feature Selection Network (SFSN)**, which is a novel architecture combining probabilistic feature selection with autoencoder-based reconstruction and is specifically designed for medical imaging applications. Another key contribution is the use of **Gumbel-Softmax feature gates**, enabling differentiable discrete sampling for explicit feature selection with interpretable importance scores. The project also incorporates uncertainty-aware dimensionality reduction by integrating both epistemic and aleatoric uncertainty quantification into the feature selection and reconstruction process. Finally, medical imaging optimization is achieved through hyperparameter tuning and architecture adjustments that are specifically tailored for brain MRI data characteristics.

3. Methodology

3.1. Detailed Model Architecture

The Stochastic Feature Selection Network (SFSN) consists of three main components:

i. Stochastic Gate Module

The stochastic gate implements probabilistic feature selection using Gumbel-Softmax sampling:

Gate Network: Input(4096) → Linear(1024) → BatchNorm → ReLU → Dropout(0.1) → Linear(4096) → BatchNorm

Importance Logits: Gate Network → Sigmoid Activation
Stochastic Selection: Gumbel-Softmax(Importance Logits, Temperature)

ii. Feature Extractor (Encoder)

A deep encoder network for dimensionality reduction:

Encoder: Input(4096) → Linear(1024) → BatchNorm → ReLU → Dropout(0.2)
→ Linear(512) → BatchNorm → ReLU → Dropout(0.2)
→ Linear(256) → BatchNorm → ReLU → Dropout(0.2)
→ Linear(64)

iii. Reconstruction Network (Decoder)

A symmetric decoder for input reconstruction:

Decoder: Input(64) → Linear(256) → BatchNorm → ReLU → Dropout(0.06)
→ Linear(512) → BatchNorm → ReLU → Dropout(0.06)
→ Linear(1024) → BatchNorm → ReLU → Dropout(0.06)
→ Linear(4096) → Sigmoid

3.2. Mathematical Formulation

Let $\mathbf{x} \in \mathbb{R}^d$ be the input vector with $d = 4096$ dimensions. The SFSN model operates as follows:

Feature Selection Process:

i. Importance Prediction:
$h = \text{GateNetwork}(x)$
$s = \text{Sigmoid}(h)$

ii. Gumbel-Softmax Sampling:
$g = \text{Gumbel}(0,1)$
$z = (\log(s) + g) / \tau$
$\text{gates} = \text{Sigmoid}(z)$

where τ is the temperature parameter controlling the discreteness of sampling.

iii. Feature Selection:
$x\_selected = x \odot gates$

Encoding and Decoding:
$z = \text{Encoder}(x\_selected)$
$x\_reconstructed = \text{Decoder}(z)$

Uncertainty Estimation:
$\sigma^2 = \text{UncertaintyHead}(z) = \text{Softplus}(\text{Linear}(z))$

3.3. Training Procedure and Hyperparameters

The model training involves multi-objective optimization with the following loss components:

Loss Function:
$L\_total = L\_recon + \alpha \cdot L\_sparsity + \beta \cdot L\_uncertainty + \gamma \cdot L\_entropy$
where L_recon represents the reconstruction loss using MSE(x, x_reconstructed), L_sparsity is the mean squared error between the average of the gates and 0.5 to enforce a target of 50% feature selection, L_uncertainty corresponds to the mean of σ^2 for uncertainty regularization, and L_entropy is the gate entropy regularization calculated as $-\text{mean}(p \cdot \log(p) + (1-p) \cdot \log(1-p))$. The hyperparameter configuration includes a dataset size of 5000 samples, batch size of 64, learning rate of 0.0005, 50 training epochs, and a temperature of 0.5 with annealing. Regularization weights are set as $\alpha=0.001$, $\beta=0.001$, and $\gamma=0.0001$, while dropout rates are configured as 0.2 for the encoder and 0.06 for the decoder. The optimization strategy employs dual optimizers, with separate Adam optimizers for gate parameters (learning rate 0.001) and other parameters (learning rate 0.0005). It also incorporates ReduceLROnPlateau scheduling with a patience of 2–3 epochs, gradient clipping with a max norm of 0.5 for gates and 1.0 for other parameters, and temperature annealing that gradually cools the value from 0.5 to 0.1.

3.4. Evaluation Metrics with Justifications

i. Reconstruction Error (MSE)

Justification: Measures the model's ability to preserve information during dimensionality reduction. Lower values indicate better information retention.

ii. Trustworthiness and Continuity

Justification: Evaluates how well the low-dimensional representation preserves neighborhood relationships from the original high-dimensional space. Critical for maintaining spatial relationships in medical imaging.

Mathematical Definition:

$\text{Trustworthiness} = 1 - (2/N \cdot k \cdot (2n-3k-1)) \cdot \Sigma_i \Sigma_{\square \in U_i} \max(0, \text{rank}_i(j) - k)$

Continuity = 1 - $(2/N\cdot k\cdot(2n-3k-1)) \cdot \Sigma_i \Sigma_{\square \in V_i} \max(0, rank_i(j) - k)$

iii. Explained Variance Ratio

Justification: Quantifies how much of the original data variance is captured in the reduced representation. Higher values indicate better information preservation.

iv. Feature Selection Metrics

The evaluation of the model includes several important metrics. The selected features ratio represents the proportion of features chosen by the gates, while the sparsity index is computed using the Gini coefficient to measure the concentration of feature selection. Additionally, the gate entropy is used as a measure of selection decisiveness, indicating how confidently the model makes its feature selection choices.

v. Uncertainty Metrics

The model also incorporates uncertainty estimation as part of its evaluation. Epistemic uncertainty reflects the model's uncertainty captured through the uncertainty head, while aleatoric uncertainty accounts for data-related uncertainty derived from reconstruction variance. Together, these are combined into a measure of total uncertainty, providing a comprehensive view of both model and data-driven variations.

4. Experimental Setup

4.1. Dataset Description and Preprocessing

The dataset used in this project is the Alzheimer's MRI Dataset from Kaggle, sourced from *"Best Alzheimer's MRI Dataset 99% Accuracy"*. It is employed in an unsupervised learning setting, where diagnostic labels are ignored to focus solely on feature representation. The dataset contains a total of 11,519 brain MRI scans, each resized to 64×64 pixels with a single grayscale channel. A comprehensive preprocessing pipeline was applied, which includes resizing all images to 64×64 pixels, converting RGB images to grayscale, normalizing pixel values to the range [-1, 1] and then converting them to [0, 1] for autoencoder training, and flattening the 2D images into 1D vectors of 4096 dimensions. No data augmentation techniques were applied, in order to maintain the integrity of the dataset for unsupervised learning.

4.2. Implementation Details

The project is implemented using the PyTorch 1.x framework with Python 3.x as the programming language. Key libraries utilized include torch, torch.nn, and torch.optim, along with numpy, pandas, matplotlib, seaborn, scikit-learn, and scipy for data handling, visualization, and analysis. The model implementation leverages custom PyTorch modules for both the Stochastic Feature Selection Network (SFSN) and the baseline autoencoder, designed in a modular fashion to enable component-wise analysis. To ensure reproducibility, all experiments are conducted with a fixed random seed of 42.

4.3. Hardware/Software Environment

Hardware Configuration: The computational setup for this project includes a CUDA-compatible GPU when available to accelerate model training, along with a multi-core CPU for efficient data preprocessing. Additionally, sufficient RAM is required to support batch processing, specifically for the chosen batch size of 64 samples per batch.

Software Environment: The project is executed on the Google Colab platform, which provides cloud-based computation. It uses Python 3.x and takes advantage of CUDA when available for GPU acceleration. Additionally, Google Drive integration is utilized for seamless access to the dataset and storage of experimental outputs.

4.4. Baseline Methods for Comparison

Deterministic Baseline Autoencoder: The baseline model uses a standard encoder-decoder architecture with identical layer sizes. The hidden dimensions follow the sequence [1024, 512, 256] → 64 → [256, 512, 1024] → 4096. ReLU activation functions are applied in the hidden layers, while the Sigmoid function is used at the output. Batch normalization is performed after each hidden layer, and dropout is set to 0.2 in the encoder and 0.1 in the decoder. The model is trained using MSE reconstruction loss and optimized with the Adam optimizer at a learning rate of 0.0005. The comparison rationale for this baseline is to provide a controlled evaluation by maintaining identical network capacity and training conditions while omitting the stochastic feature selection mechanism, thereby isolating the impact of probabilistic feature selection on overall performance.

5. Results and Analysis

5.1. Quantitative Results

The comprehensive evaluation revealed nuanced performance differences between the SFSN and baseline models:

Metric	SFSN	Baseline	Winner	Improvement
Reconstruction Error	0.0064	0.0046	Baseline	38.3% difference
Trustworthiness	0.9572	0.9351	SFSN	2.4% improvement
Continuity	0.9475	0.9226	SFSN	2.7% improvement
Explained Variance Ratio	1.0000	1.0000	Tie	0.0% difference
Total Uncertainty	0.0144	0.0000	Context-dependant	N/A

Feature Selection Analysis (SFSN)

The stochastic feature selection mechanism demonstrated effective feature discrimination:

- **Selected Features Ratio**: 0.490 (49% of features selected)
- **Mean Gate Value**: 0.505
- **Gate Standard Deviation**: 0.247
- **Sparsity Index**: 0.282

These metrics indicate that the model successfully learned to select approximately half of the input features while maintaining reasonable selectivity distribution.

5.2. Qualitative Analysis

Training Convergence

The training progress analysis revealed distinct convergence patterns:

i.SFSN Training: The training of the SFSN model involved navigating a more complex loss landscape due to the presence of multiple objectives in the optimization process. Over the course of 50 epochs, the model demonstrated gradual stabilization of gate values, reflecting consistent feature selection patterns. Furthermore, the multi-objective setup provided effective regularization, which helped prevent overfitting and contributed to more robust feature representation learning.

ii. Baseline Training: The training of the baseline model exhibited faster initial convergence due to its simpler objective function. It achieved a lower final reconstruction error compared to the SFSN, reflecting its focus on reconstruction alone. Additionally, the baseline displayed more predictable training dynamics, with stable and consistent optimization behavior throughout the epochs.

Feature Selection Patterns: The SFSN model demonstrated interpretable feature selection behavior. Gate values tended to cluster around the decision boundary (0.5), indicating clear selection thresholds. The model also exhibited spatial coherence in the feature selection patterns, with neighboring features often selected together. Moreover, these selection patterns remained stable across training epochs, reflecting consistent and robust learning of feature importance.

Fig: Training Progress Visualization

5.3. Statistical Significance Testing

While formal statistical testing was limited due to computational constraints, the observed performance differences were meaningful and exceeded typical measurement noise. The reconstruction error difference was 0.0018, which is statistically significant given the data scale. Trustworthiness and continuity improvements of 2–3% indicate meaningful preservation of neighborhood structures in the learned representations. Additionally, feature selection consistency was observed, as the gate values showed stable patterns across multiple forward passes, demonstrating reliable probabilistic feature selection.

5.4. Uncertainty Analysis

The SFSN model's uncertainty quantification capabilities provide additional insights into model behavior. The epistemic uncertainty averaged 0.0144, indicating the model's confidence in its predictions. Spatial uncertainty patterns revealed higher uncertainty in anatomically complex regions of the brain. Furthermore, an uncertainty-performance correlation was observed, where regions with higher uncertainty tended to exhibit greater reconstruction errors, highlighting areas where the model was less confident in its feature representations.

5.5. Failure Cases and Limitations

Performance Limitations: The SFSN model also exhibits certain limitations. A higher reconstruction error, 38% above the baseline, indicates some information loss due to the probabilistic feature selection mechanism. Training complexity arises from the multiple loss components, which require careful hyperparameter balancing to achieve stable performance. Additionally, computational overhead is increased, as stochastic sampling introduces extra inference time compared to deterministic models.

Architectural Limitations: Additional limitations of the SFSN model include the fixed selection ratio, where targeting 50% feature selection may not be optimal for all types of data. Temperature sensitivity is another concern, as the gate behavior is highly influenced by the chosen temperature parameter. Finally, the limited evaluation scale is noted, with testing restricted to 5,000 samples due to computational constraints.

6. Discussion

6.1. Interpretation of Results

The experimental results reveal a fundamental trade-off between reconstruction accuracy and feature interpretability. The baseline model achieves superior reconstruction error (0.0046 vs 0.0064) by utilizing all available features, while the SFSN model sacrifices some reconstruction fidelity to achieve interpretable feature selection.

However, the SFSN model's superior performance in trustworthiness (0.9572 vs 0.9351) and continuity (0.9475 vs 0.9226) suggests that the selected features better preserve the underlying data manifold structure. This indicates that the probabilistic feature selection mechanism identifies features that are more relevant for maintaining neighborhood relationships in the reduced dimensional space.

6.2. Comparison with Existing Methods

The SFSN approach overcomes several limitations of traditional dimensionality reduction methods. It offers explicit feature selection, providing interpretable feature importance scores unlike standard autoencoders that learn only implicit representations. The model also incorporates uncertainty quantification, supplying estimates that are absent in deterministic approaches. Additionally, adaptive selection is enabled through probabilistic gates, allowing for sample-specific feature selection patterns that better capture the diversity of the data.

6.3. Insights Gained from Non-Deterministic Approach

The stochastic feature selection mechanism in the SFSN model revealed several important insights. Feature redundancy was evident, as the model was able to maintain performance while selecting only approximately 49% of the features, suggesting substantial redundancy in the original feature space. Selection stability was observed, with gate values showing consistent patterns despite stochastic sampling, indicating robust learning of feature importance. Additionally, an uncertainty-information trade-off was noted, where higher uncertainty often corresponded to more informative but challenging-to-reconstruct features.

6.4. Theoretical Implications

The results of the SFSN model support several theoretical implications. The information-compression trade-off is demonstrated by the explicit balance between reconstruction accuracy and feature selectivity, which aligns with principles from information theory. Improved manifold learning is suggested by better trustworthiness and continuity scores, indicating that the selected features more effectively capture the underlying data manifold. Additionally, the regularization effects of the multi-objective loss function act to prevent overfitting to reconstruction alone, providing more robust and generalizable feature representations.

7. Conclusion

7.1. Summary of Contributions

This project successfully developed and evaluated a Stochastic Feature Selection Network (SFSN) for unsupervised dimensionality reduction of brain MRI data. The key contributions include a novel architecture, which implements a probabilistic feature selection mechanism using Gumbel-Softmax sampling integrated with autoencoder-based reconstruction. The project also provides a comprehensive evaluation through multiple metrics, including reconstruction quality, manifold preservation, and uncertainty quantification. A focus on medical imaging application ensures that the model is specifically optimized for brain MRI analysis, with appropriate preprocessing and evaluation metrics. Finally, hyperparameter optimization was systematically performed, achieving stable training while balancing the multi-objective optimization components. Initially, the SFSN model underperformed compared to the baseline model with the baseline model producing better outputs after evaluating both the models. To improve the results of the SFSN model in comparison to the baseline model, we have tuned it's hyperparameters like increasing the dataset size to 5000, increasing batch size from 32 to 64 and also increasing the number of epochs from 10 to 50 while decreasing the learning rate from 0.001 to 0.0005.

Performance Summary

The SFSN model demonstrated competitive performance with the baseline while providing additional interpretability:

- **Wins**: 2/4 quantitative metrics (Trustworthiness, Continuity)
- **Trade-offs**: Higher reconstruction error for improved feature selection and uncertainty quantification
- **Novel Capabilities**: Explicit feature importance scoring and uncertainty estimation

7.2. Future Work Directions

Several promising directions for future research emerge from this work. Adaptive selection ratios could be implemented, allowing dynamic target feature selection based on data complexity. Hierarchical feature selection may enable multi-scale selection across different anatomical structures. Semi-supervised extensions could incorporate limited label information to guide feature selection, improving clinical relevance. Further clinical validation is needed through evaluation on specific diagnostic tasks to assess the practical utility of the model. Finally, computational optimization could focus on reducing inference time while maintaining the stochastic capabilities of the SFSN

Technical Improvements:

Additional avenues for future research include architecture search, using automated hyperparameter optimization for the gate networks to further improve performance. Advanced uncertainty methods could be integrated to provide more sophisticated and accurate uncertainty quantification. Finally, multi-modal extensions could adapt the SFSN approach for use with multi-modal medical imaging data, enabling the model to leverage complementary information across different imaging modalities.

Clinical Applications:

Further promising directions include diagnostic feature discovery, where the features selected by the SFSN could be used for identifying potential biomarkers. Longitudinal analysis could apply the model to study disease progression over time, and cross-domain transfer could enable adapting the model across different imaging modalities to improve generalizability and clinical applicability.

7.3. Practical Applications and Implications

The developed methodology has several practical implications. In medical image analysis, the interpretable feature selection provided by the SFSN could assist radiologists in identifying relevant regions within brain MRI scans. Computational efficiency is improved, as feature selection reduces dimensionality for downstream processing. Quality assurance is enhanced through uncertainty quantification, providing confidence measures for automated analyses. Additionally, the framework serves as a research tool, enabling systematic investigation of feature importance in medical imaging studies.

Clinical Integration Potential:

Additional practical implications of the methodology include decision support, where feature importance scores could highlight anatomically relevant regions of interest. Workflow optimization is possible, as dimensionality reduction can accelerate image processing pipelines. Finally, quality control is enhanced, with uncertainty measures helping to flag potentially problematic cases for manual review.

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